CS and PS

Reference: Microeconomics by Mankiw (8e)

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Today's Topic

- Today we will understand the concepts of CS and PS in some detail.
- This will help us also understand what we consider efficient and welfare maximizing.
- We do not delve into this topic at a higher level, so please read Mankiw.

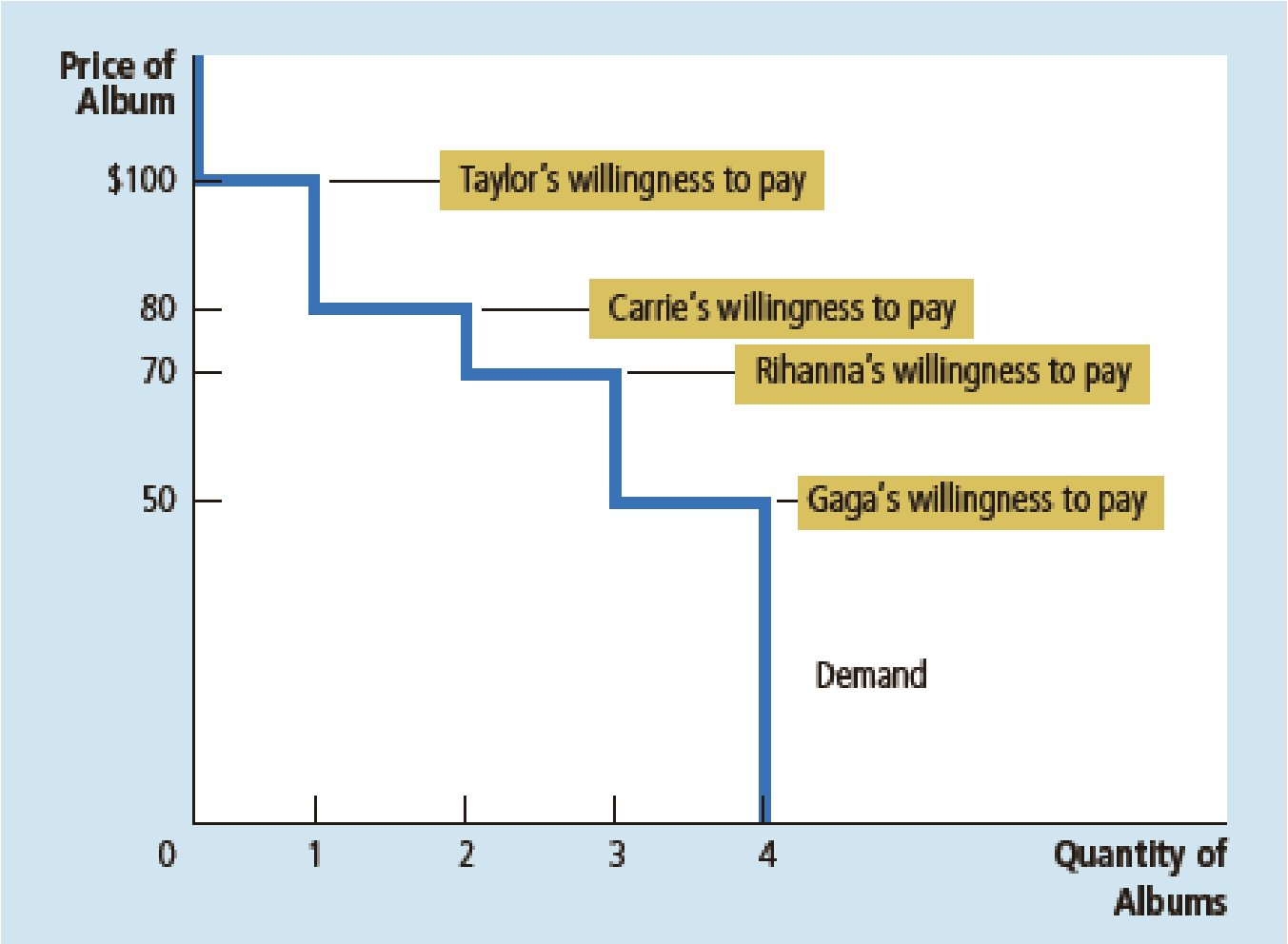
What is CS?

- Now that we have seen utility, CS is quite simply the utility after you buy a good, subtracting the price paid.
- In a similar sense, PS is the profit.
- CS can be for an individual or for everyone. It makes more sense to think of it for everyone.
- Same for PS

WTPs

Buyer	Willingness to Pay
Taylor	$100
Carrie	80
Rihanna	70
Gaga	50

WTPs Graphically

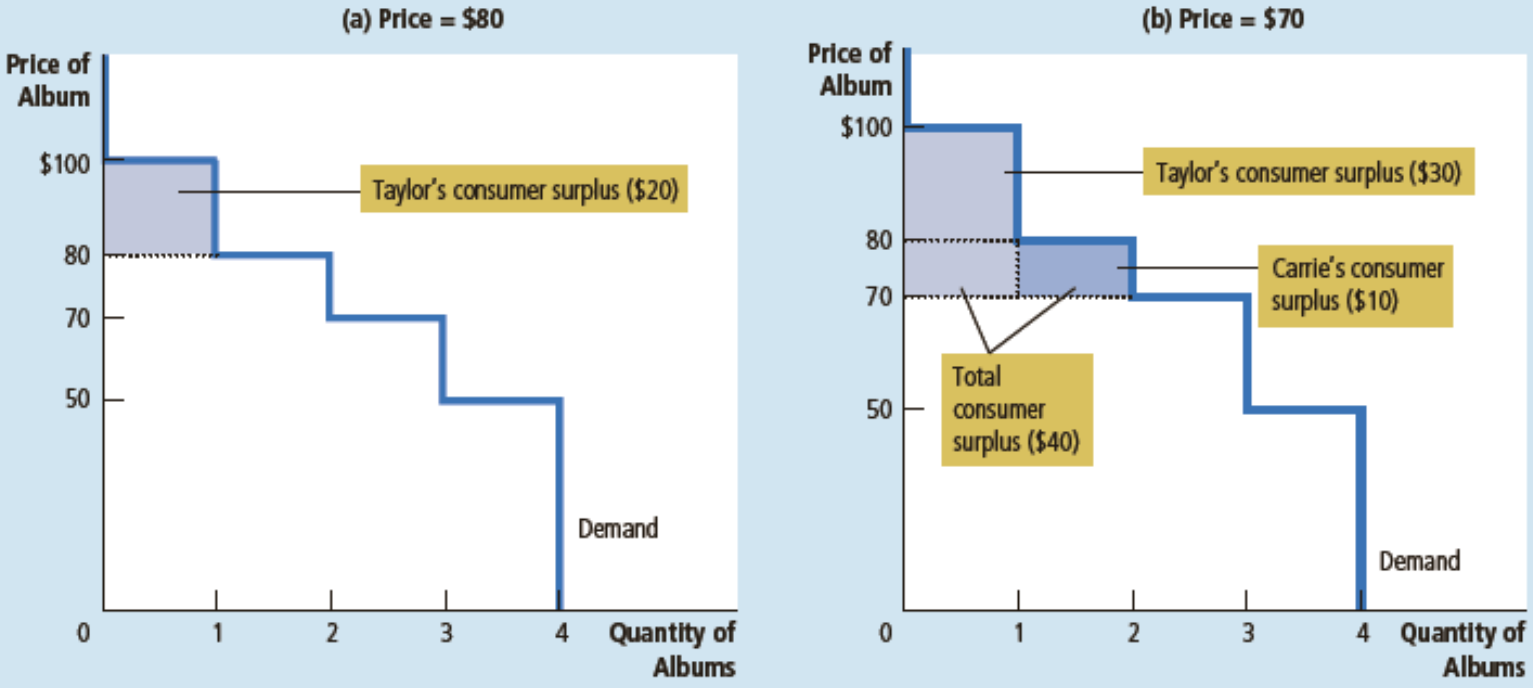


How CS looks

In panel (a), the price of the good is $80 and the consumer surplus is $20.
In panel (b), the price of the good is $70 and the consumer surplus is $40.

FIGURE 2

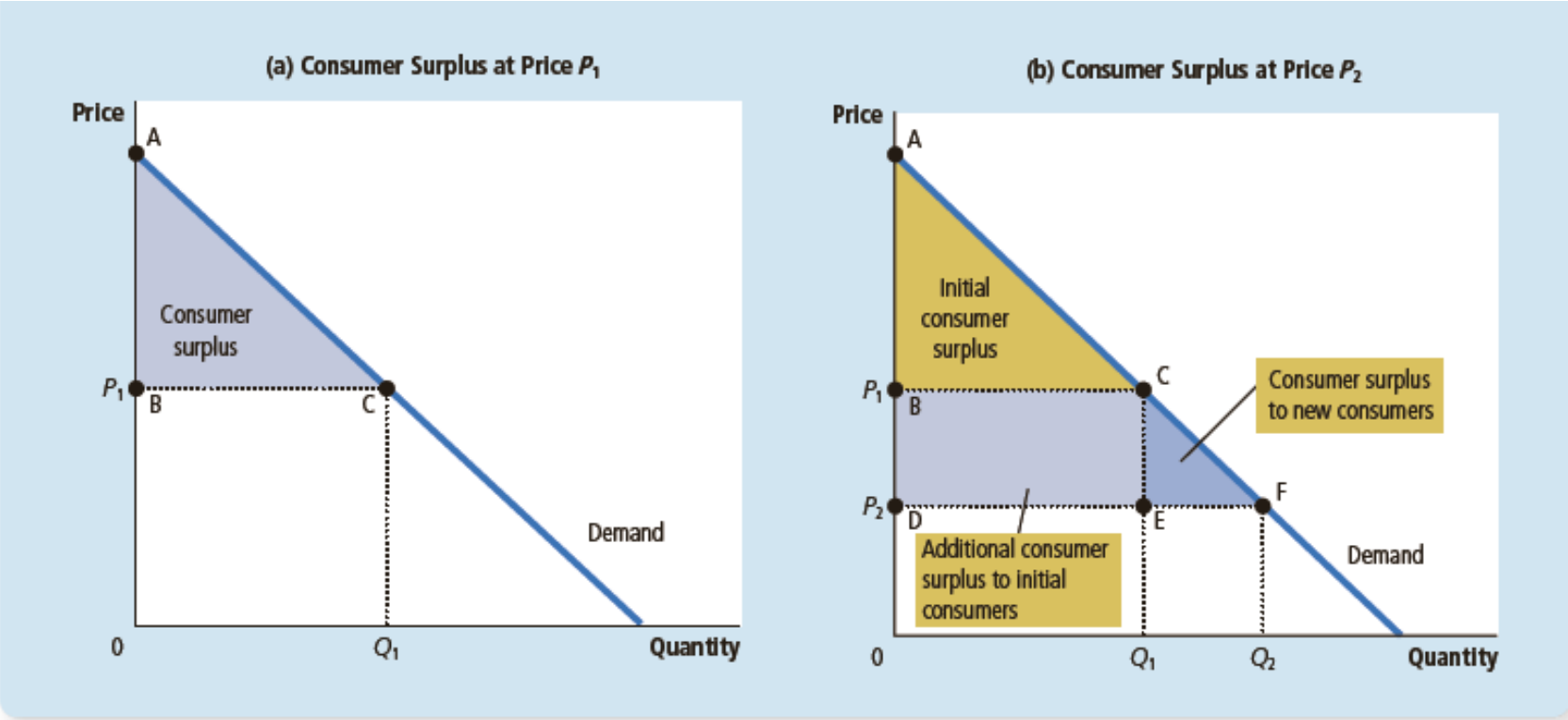
Measuring Consumer Surplus with the Demand Curve



CS and Price

- We cannot calculate CS without a price. However, once we have a price, we can calculate CS independent of the shape of the Supply curve.
- CS is one of the most important metrics calculated for many purposes.
- For example, most guidelines analyse CS while approving mergers.

How change in price looks



PS

- PS is easier to imagine.
- Whether or not policy makers care about PS is a little more context specific.
- They may care for farmers, but not for huge industrialists (especially in the context of competition policy)

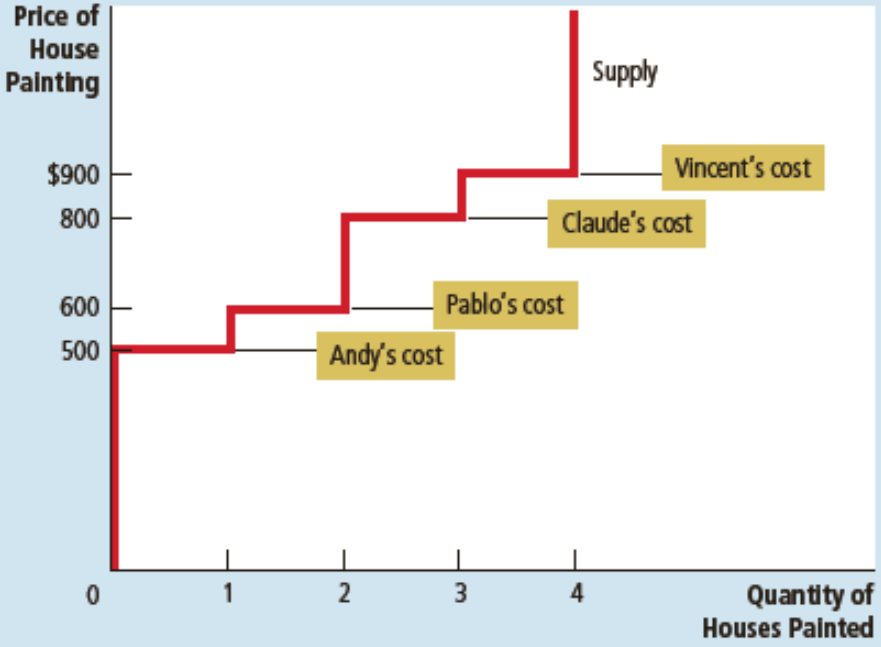
How PS looks

FIGURE 4

The Supply Schedule and the Supply Curve

The table shows the supply schedule for the sellers (listed in Table 2) of painting services. The graph shows the corresponding supply curve. Note that the height of the supply curve reflects the sellers' costs.

Price	Sellers	Quantity Demanded
$900 or more	Vincent, Claude, Pablo, Andy	4
$800 to $900	Claude, Pablo, Andy	3
$600 to $800	Pablo, Andy	2
$500 to $600	Andy	1
Less than $500	None	0

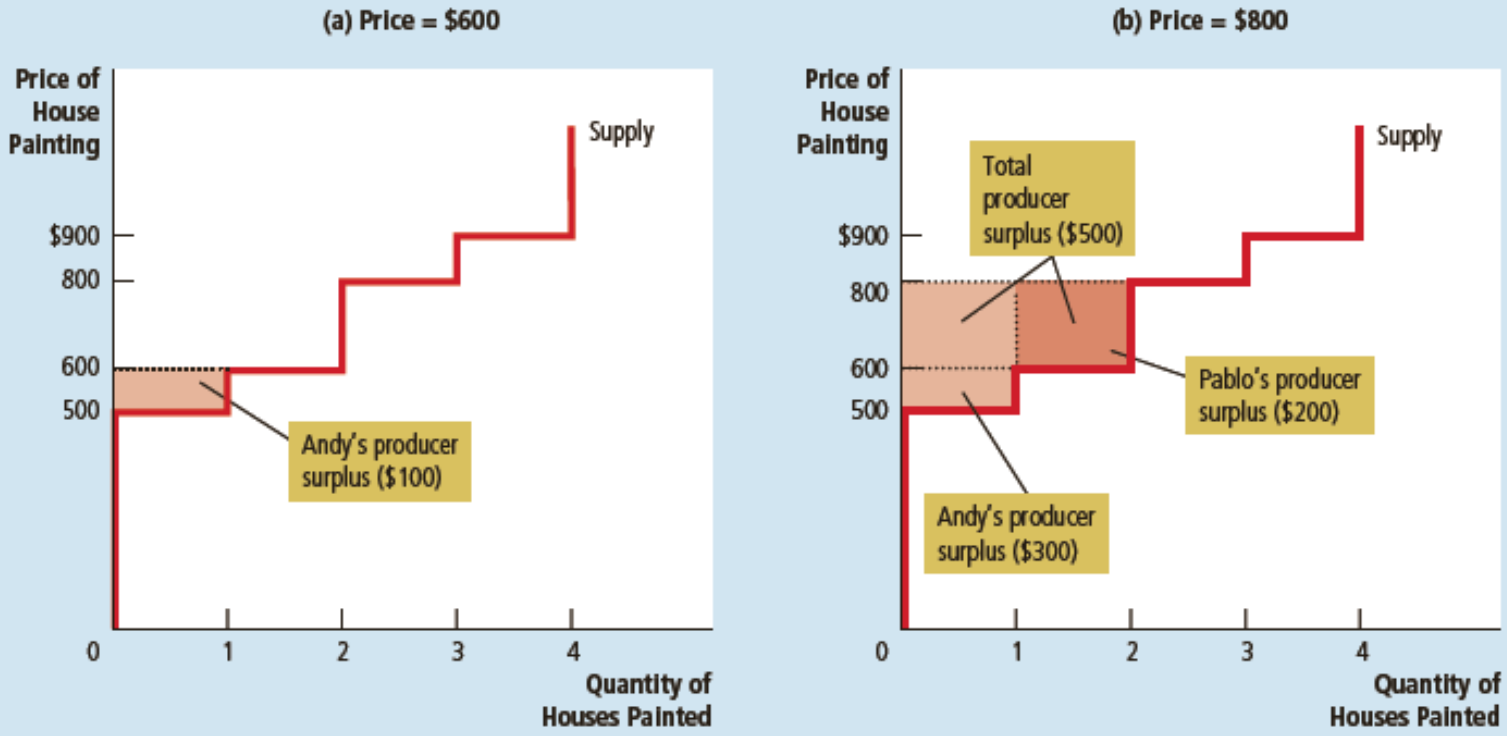


How PS looks

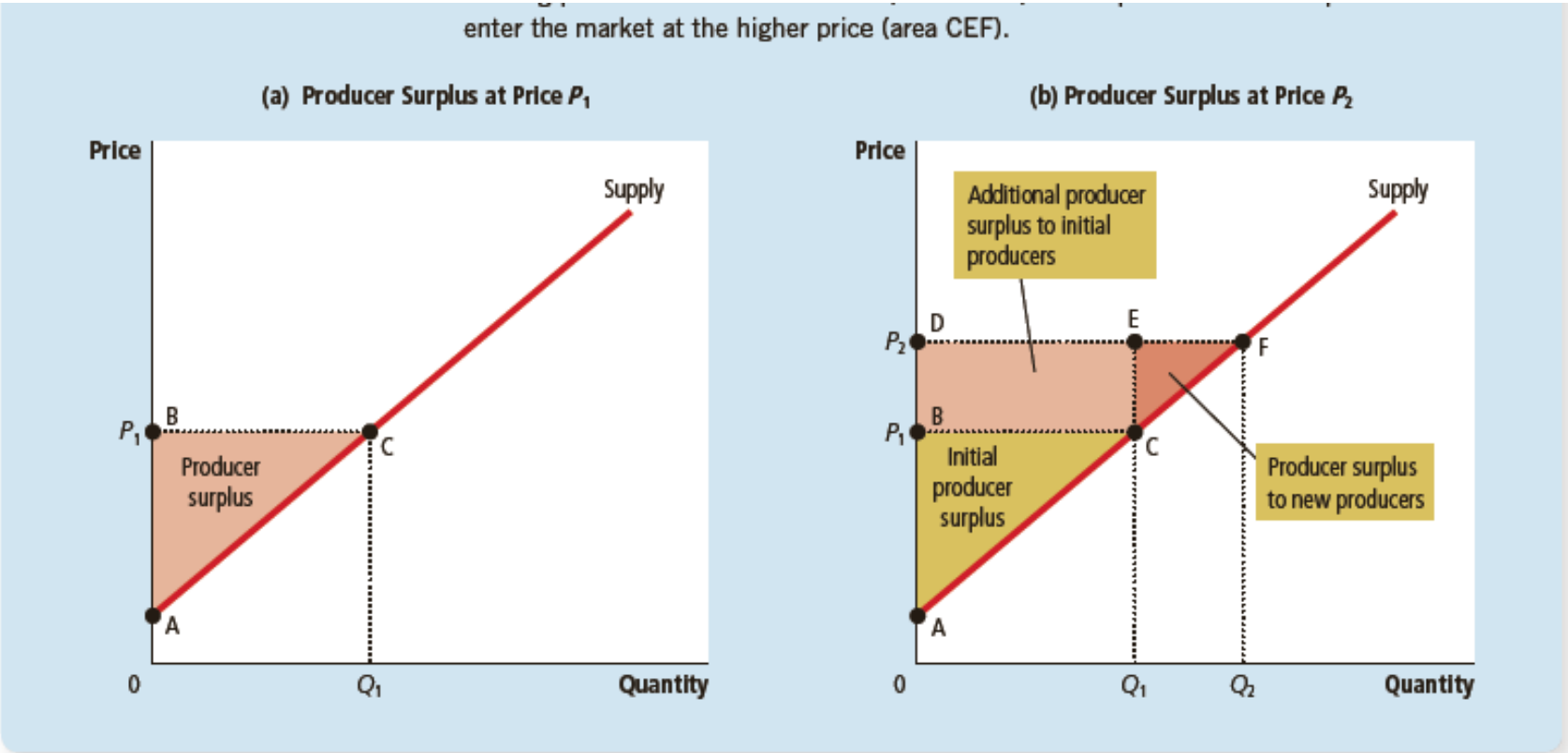
In panel (a), the price of the good is $600 and the producer surplus is $100.
In panel (b), the price of the good is $800 and the producer surplus is $500.

FIGURE 5

Measuring Producer Surplus with the Supply Curve



PS and price


enter the market at the higher price (area CEF).
(a) Producer Surplus at Price P_1
Price
Supply
P_1
B
C
Producer surplus
A
0
Q_1
Quantity
(b) Producer Surplus at Price P_2
Price
Additional producer surplus to initial producers
Supply
P_2
D
E
F
P_1
B
C
Initial producer surplus
Producer surplus to new producers
A
0
Q_1
Q_2
Quantity

Welfare

- In most basic terms, Welfare is fiven by CS+PS.
- Graphically, it is the sum of the areas.
- The general idea is with large number of buyers and sellers, we get a welfare maximizing outcome or efficient market.

Inefficient Markets

- Inefficient markets can happen.
- One major reason is market power. Too few firms (monopoly or oligopoly).
- Another is externalities. This requires a little more thinking.
- In general, these two factors are critical.
- What is demand is non-linear